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Multistage Constant-Current Charging Method for Li-ion Batteries

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***Abstract*—This paper proposes a unique method to find the optimal charge pattern (OCP) of multistage constant-current charging (MSCCC) method by using RC based equivalent circuit model for Li-Ion batteries. Both 3 and 5 stages based MSCCC methods are discussed. The method of finding the equivalent circuit model and optimal charge pattern by equations are also explained. Experimental results with a 2.6 Ah Li-ion battery are used to validate the proposed method. The proposed method is unique in that it can charge the battery upto 100 % capacity in a shorter time. Compared with conventional CC-CV method, the charging time (CT) and the charging efficiency by the proposed method are improved by approximately 12% and 0.54% respectively.**

Keywords—* *Fast charging; Multi-stage charging; Li-Ion battery; Optimal charge pattern

I. Introduction

Li-Ion batteries have become very popular in portable electronic appliances, aerospace, aircraft, transportation vehicles, and energy backup for renewable power systems due to their several appealing features of high power densities, wide operating temperature range, long life cycles, no memory effect and low self-discharge rate. However Li-ion batteries are also very sensitive to deep discharge or over charge, temperature and charge/discharge currents. Therefore for the applications of lithium ion batteries, a well-designed battery charger is required to get high battery performance and lifespan. The charger should have the qualities to maximize the charging capacity, shorten the charge time (CT) and lengthen the cycle life of the battery.

Fast battery charging without affecting its electro-chemistry has always been a focus of charger development techniques. Mostly a conventional CC-CV method is used to charge Li-ion batteries. In CC mode, a higher constant current is used to charge the battery until its predefined cut-off voltage value (typically 4.2V) has reached. In CV mode, same cut-off voltage is applied constantly until the current is decreased to its cut-off value (typically 0.01C) and after that charging is stopped. If higher current is used in CC mode then the CV mode will prolong the charging time and also has adverse effects on the charging efficiency and charging capacity. Hence the fast CC-CV charge method is not feasible.

Several studies have investigated different techniques and topologies to reduce the charging time (CT) of the battery. Multistage constant current charging (MSCCC) method is one of the popular methods for fast charging. Different optimization techniques such as Ant Colony System Algorithm, Particle Swarm Optimization (PSO), Taguchi Approach and Fuzzy Logics are used to find the optimal charge pattern (OCP) of multistage constant-current charging (MSCCC) method [3]-[7]. First of all different charge patterns are applied on the batteries and their respective charging times (CT) and discharged capacities are recorded. After that above mentioned optimization techniques are used to find the optimal charge pattern (OCP). There are some drawbacks in the execution of MSCCC method which can be observed by carefully analyzing the previous work. The first drawback of this method is its tedious procedure to find OCP which sometimes requires hundreds of experiments. The second one is its lack of ability to charge the battery upto its full capacity in a short period of time as compared to CC-CV method. Thus it is not suitable for those applications which requires charging the battery up to its 100 % capacity. Hence, all the previous research work only focus on charging the battery up to 95 % of its capacity and compare the charge time with that by CC-CV method (100% charged). However, it is not fair since the comparison must be performed for the times taken to charge the same capacity of the battery by both methods [3]-[7].

This paper introduces a new method to find the OCP of MSCCC method by using a simple RC based equivalent circuit model. Followings are the advantages of proposed method:

1. The proposed method can be used to find the OCP by doing only one experiment on the battery as compared to hundreds of experiments required by the conventional MSCCC methods.
2. The proposed method can charge the battery to any required capacity within shorter time as compared to CC-CV method. If battery is charged upto its 100 % capacity then the charging time of proposed method is 12 % less than that of CC-CV method.

II. Proposed Method

A. Comparison of Various Multistage Charge Methods

Reference [1] examines and tests the effects of different number of current levels (stages) on the charging time (CT) and charging efficiency. According to the experimental results, the difference between applying a different number of charging current levels is almost negligible when the number of charging current levels is more than 5.This paper presented a comparison of 3 stages constant-current charging (3SCCC) and 5 stages constant-current charging (5SCCC) methods. It was observed that same capacity can be charged by using either 3SCCC or 5SCCC methods but the charging time (CT) of 3SCCC method is always higher than that of 5SCCC method. Therefore 5SCCC method is preferred for rapid charging as it can acquire more charge in shorter charging time. All the details about 3SCCC and 5SCCC methods can be found in the Section III.

B. Calculation of Charging Time and Charging Capacity

A Li-Ion battery can be modelled by using an RC based equivalent circuit model as shown in the Figure 1.

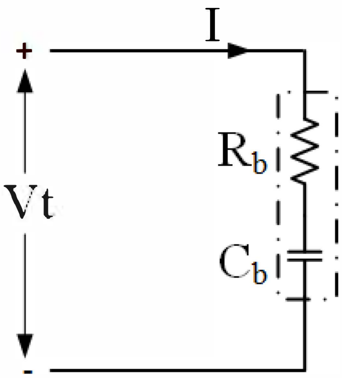

Fig. 1. RC Equivalent Circuit Model of Li-Ion Battery

R_b and C_b represents the internal resistance and bulk capacitance of the battery respectively. While V_t represents the terminal voltage of the battery. When a constant current I is applied across the battery its terminal voltage can be calculated as:

$$V_t = V_{C_b} + V_{R_b} \tag{1}$$

$$V_{C_b} = \frac{1}{C_b}\int_{t_0}^{t} I(\tau)d\tau + V_{in} \text{ and } V_{R_b} = I \times R_b \tag{2}$$

where V_{Cb} is the voltage across the battery when a certain current I is applied across it and V_{in} is the initial voltage across the battery when the battery is in resting condition or it can be called as the initial OCV of the battery. Combining the above two equations,

$$V_t = \frac{1}{C_b}\int_{t_0}^{t} I(\tau)d\tau + V_{in} + V_{R_b} \tag{3}$$

As the final cut-off voltage of a battery is always known so the time taken by the battery to reach that cut-off voltage when current I is applied can be calculated as:

$$\Delta t = \frac{1}{I}\left(V_t - V_{in} - V_{R_b}\right) \times C_b \tag{4}$$

If five continuous constant current pulses are applied across the battery then the total time taken by each pulse will be given as:

$$T = \Delta t_1 + \Delta t_2 + \Delta t_3 + \Delta t_4 + \Delta t_5 \tag{5}$$

where

$$\Delta t_x = \frac{1}{I_x}\left(V_t - V_{in}^x - V_{R_b}^x\right) \times C_b \tag{6}$$

$$V_{in}^1 = OCV \tag{7}$$

$$V_{in}^x = V_t - I_{x-1} \times R_b \qquad x = 2,3,\ldots,5 \tag{8}$$

$$V_{R_b}^x = I_x \times R_b \qquad x = 1,2,3,\ldots,5 \tag{9}$$

T represents the total time of five stages constant current charging method and x represents the stage number of the charging method. Similarly (4) can be used to calculate the Ah charged by the battery when I current is applied across it and its voltage changes from OCV to final cut-off voltage as:

$$\Delta Ah = I \times \Delta t = \frac{1}{3600}\left(V_t - V_{in} - V_{R_b}\right) \times C_b \tag{10}$$

The total Ah charged by all of the five stages will be given as:

$$Ah = \Delta Ah_1 + \Delta Ah_2 + \Delta Ah_3 + \Delta Ah_4 + \Delta Ah_5 \tag{11}$$

$$\Delta Ah_x = \frac{1}{3600}\left(V_t - V_{in}^x - V_{R_b}^x\right) \times C_b \tag{12}$$

where V_{in}^x and $V_{R_b}^x$ can be calculated by using (7)-(9).

The above analysis can be used to calculate the total charging time and charged capacity of the battery for any number of stages used in MSCCC method.

C. Derivation of Optimal Charge Pattern

It can be observed from (11) that the charged Ah capacity in MSCCC method depends only on the value of the last stage current if the value of first stage current is taken as constant. First stage current I_1 is always selected to be the highest allowed current in the CC mode by the manufacturer of the battery. For example, if two 5SCCC profiles e.g. [2.6,2.1,1.5,0.9,0.5] Amps and [2.6,2.2,1.8,0.7,0.5] Amps are applied on a Li-Ion battery then Ah capacity charged by both of the profiles will approximately be the same because I_1 and I_5 are same for both of them but their charging time will be different depending upon the values of I_2, I_3 and I_4. The experimental proof of this phenomenon is given in section III. So if I_1 is constant then charged Ah capacity by a 5SCCC profile only depends upon the value of I_5 while the charging time depends upon the values of I_2, I_3 and I_4. Same rule is also applicable to 3SCCC method.

To find the equations for optimal charge pattern, 3SCCC method is considered for simplicity. Suppose that it is required to charge a li-ion battery with 3SCCC method and any arbitrary values of I_1 & I_3 are considered for the analysis. The current I_1 is the maximum current used in the CC mode and I_3 is decided based on the required charged capacity of the battery. The lesser the value of I_3, the higher the value of charged capacity and vice versa. The total charging time taken by the three stages and derivative of total time w.r.t. I_2 can be given as:

$$T = \Delta t_1 + \Delta t_2 + \Delta t_3 \tag{13}$$

$$\frac{dT}{dI_2} = \frac{-I_1 R_b C_b}{I_2^2} + \frac{R_b C_b}{I_3} \quad (14)$$

By putting derivative equals to zero and solving for I_2, (15) is obtained.

$$I_2 = \sqrt{I_1 I_3} \quad (15)$$

Fig. 2 shows the graph of total charging time and value of I_2 current while using some constant arbitrary values of I_1, I_3, V_t, V_{in}, R_b and C_b. It is clearly depicted in the Fig.2 that charging time is minimum when I_2 has the same value as calculated by (15). It is also verified that the value of I_2 is independent of the values of R_b and C_b.

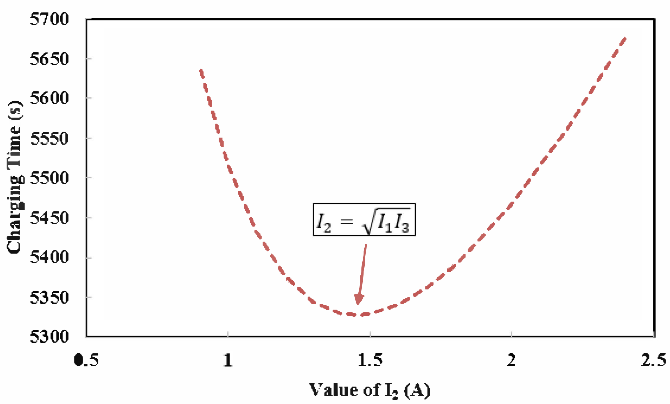

Fig. 2. Charging Time vs value of I_2

Current I_2 only depends upon the values of currents of previous and next stages. Similarly for 5SCCC method if I_1 and I_5 are known then using the previous procedure, each stage current can be calculated as in (16) to (19).

$$I_2 = \sqrt{I_1 I_3} \quad (16)$$

$$I_3 = \sqrt{I_2 I_4} \quad (17)$$

$$I_4 = \sqrt{I_3 I_5} \quad (18)$$

where I_3 can also be shown as:

$$I_3 = \sqrt{I_1 I_5} \quad (19)$$

If OCV and RC equivalent circuit model of a battery are known then the optimal charging pattern can be found which can charge the battery up to any certain capacity in shorter charging time. So there is no need for performing hundreds of experiments on a battery to find its optimal charge pattern as it can be easily found by using the above equations.

D. Consideration of the Cut-off Voltage

Typically most of the Li-Ion batteries have a cut-off voltage of 4.2±0.05 V but normally 4.2V is used for safety purposes. Reference [2] has successfully tried 4.25V for the CV mode to get shorter charging time and charging energy. Using 4.22V as cut-off voltage for each stage of MSCCC method will not have any negative effect on the battery as the period of application of this high cut-off voltage is very small as compared to CV mode of CC-CV charging method in which constant voltage is applied for a long period of time and can be dangerous for battery's life. Furthermore a high cut-off voltage value will help in charging the battery up to 100 % capacity by the proposed method. So 4.22V is selected as cut-off voltage for MSCCC method.

III. Experimental Setup and Results

In the experimental setup a bipolar power supply is used for charging/discharging the battery. A sensing circuit and a Labview myDAQ are used for monitoring and recording the results in the computer. A brand new 2600 mAh (3.6V) lithium ion battery from LG Corporation is used in the experiments. The battery is only selected when it passed screening test with acceptable OCV, internal impedance and rated capacity. Similarly battery is placed in a chamber to maintain its temperature at 25℃. To perform the comparison of charged/discharged capacities with MSCCC method and CC-CV method, normalized discharge capacity (NDC) and normalized charge capacity (NCC) indices are introduced as:

$$NDC = \frac{C_{d-Multi}}{C_{d-CCCV}} \quad (20)$$

$$NCC = \frac{C_{c-Multi}}{C_{d-CCCV}} \quad (21)$$

where $C_{c\text{-}Multi}$ & $C_{d\text{-}Multi}$ denotes the charged and discharged capacities by MSCCC method respectively and $C_{d\text{-}CC\text{-}CV}$ is the discharged capacity of the battery by CC-CV method.

In experimental results both NDC and NCC terms are used for comparison purpose. After screening the battery, a randomly generated 5SCCC pattern is applied to the battery. Then R_b & C_b values are calculated by using (3) & (4) at different stages and their average is calculated. I_1 is selected to be 1C (2.6 A) as provided by the manufacturer of the battery. The calculated values of R_b & C_b are 0.068 Ω and 11030 F respectively while the value of V_{in} is 3.35 V.

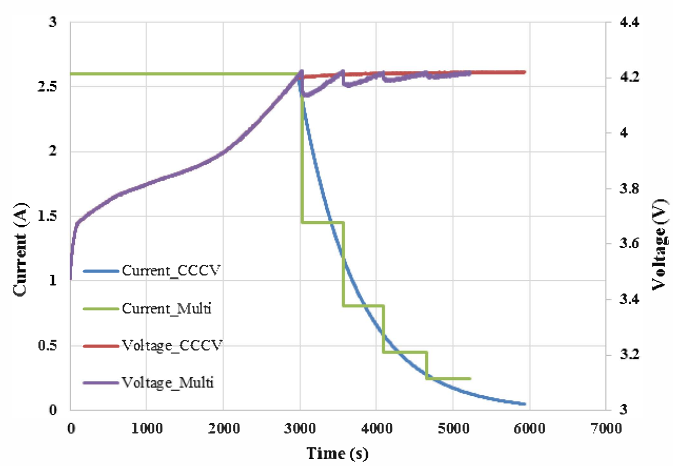

Fig. 3. Comparison of CC-CV method and optimal charge pattern of proposed method

A comparison of 5SCCC and 3SCCC methods is shown in table 1 and 2. It is clear from the results that a simple *RC* model of a li-ion battery works perfectly for calculating charging time and charged capacity by MSCCC method. The maximum difference in calculated and observed charging time and charging capacities is 40 sec and 0.5 %, respectively, which are negligible. The NDC depends upon the complete charge pattern as every charge pattern has different charging efficiency. By using 2.6A as I_1 and 0.25A as I_5, the battery can be charged upto its 100% capacity. The optimal charge pattern for the battery can be calculated as [2.6, 1.45, 0.81, 0.45, 0.25] Amps. In table 3, different charging patterns are applied in which I_1 and I_5 are same. It is proved that only the optimal charging pattern can charge the battery in shorter time and with higher charging efficiency as compared to other patterns.

To do the comparison with CC-CV method, same starting current (2.6 A) is used for both methods. The CV mode is finished when current reaches the value of 0.05A. Charging time and charging efficiency for CC-CV method are 5930 seconds (98.83 minutes) and 98.83%, respectively, while charging time and charging efficiency for optimal charge pattern of 5SCCC method are 5224 seconds (87.06 minutes) and 99.4%, respectively. So proposed method can charge the battery upto its 100 % capacity with 12 % less charging time and 0.54% higher charging efficiency as compared to CC-CV method. Fig. 3 shows the charge plots by both CC-CV method and proposed method.

TABLE I
EXPERIMENTAL & CALCULATED RESULTS OF 5SCCC METHOD

Current Pattern					**Results from Experiments**				**Results from Calculations**		**Difference in Experiment and Calculation**	
I_1	I_2	I_3	I_4	I_5	CT (s)	NCC (%)	NDC (%)	Efficiency (%)	CT (s)	NCC (%)	CT (s)	CDC (%)
2.6	1.8	1.2	0.8	0.4	4797	98.65	97.62	98.95	4774	98.52	23	0.13
2.6	2.1	1.8	1.1	0.6	4339	96.47	95.27	98.75	4347	96.93	8	0.46
2.6	2.1	1.4	0.9	0.5	4530	97.54	96.36	98.79	4511	97.72	19	0.18
2.6	2.2	1.5	1	0.45	4695	97.98	96.58	98.57	4719	98.12	24	0.14
2.6	1.5	1	0.8	0.3	5328	99.48	98.17	98.68	5303	99.31	25	0.17

TABLE II
EXPERIMENTAL & CALCULATED RESULTS OF 3SCCC METHOD

Current Pattern			**Results from Experiments**				**Results from Calculations**		**Difference in Experiment and Calculation**	
I_1	I_2	I_3	CT (s)	NCC (%)	NDC (%)	Efficiency (%)	CT (s)	NCC (%)	CT (s)	CDC (%)
2.6	1.2	0.4	5346	98.45	97.24	98.77	5316	98.52	30	0.07
2.6	1.8	0.6	4787	97.07	95.86	98.75	4774	96.93	12	0.14
2.6	1.21	0.5	4902	97.48	96.23	98.71	4867	97.72	35	0.24
2.6	1.7	0.45	5450	97.84	96.31	98.43	5421	98.12	29	0.28
2.6	1.2	0.3	6032	99.11	98.01	98.89	6066	99.31	34	0.2

TABLE III
EXPERIMENTAL RESULTS FOR THE COMPARISON OF OPTIMAL CHARGE PATTERN

Current Pattern					**Results from Experiments**			
I_1	I_2	I_3	I_4	I_5	CT (s)	NCC (%)	NDC (%)	Efficiency (%)
2.6	1.8	1	0.5	0.25	5364	100.66	99.62	98.96
2.6	1.5	0.9	0.6	0.25	5469	100.69	99.23	98.55
2.6	1.45	0.81	0.5	0.25	5224	100.55	99.95	99.4
2.6	2.2	1.8	1	0.25	5970	100.83	98.89	98.07
2.6	2	1.5	0.8	0.25	5663	100.82	99.13	98.32

IV. CONCLUSION

This paper has proposed a method to find the optimal charge pattern of MSCCC method for charging Li-ion batteries. Optimal charge pattern is calculated from the RC model of the battery. The experimental results validates that the optimal charge pattern is independent of R_b & C_b values. Each current value depends on the previous and next current values as shown in Eq. (15)-(19). The proposed method can charge the battery upto its 100 % capacity by 12% less charging time with 0.54 % higher charging efficiency as compared to the conventional CC-CV method.

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